XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	CST103			
Course Name:	Programming Language C			
Question Paper Setter:	Hamid Tahaei			
Academic Session:	2021/04	Question Paper:	A ☑	B ☐
Total No. of Pages:	4	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 2 Section(s):

   Section A: 4 question(s)
   Section B: 4 question(s)

   Answer all questions.

2. Please follow the requirement of each section and type in all the corresponding answers on the downloadable answer book prosvided.

3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Section A (50 marks)

Read Code

Question 1 (10 marks)

Predict the results that source code will produce when executed. Show the output by writing the result.

```
int main()
{
   for (int i = 1; i <= 5; i++) {
       for (int j = 1; j <= i; ++j)
         printf("%d ", i-j+1);
    printf("\n");
   }

   return 0;
}
```

Question 2 (10 marks)

Predict the results that source code will produce when executed. Show the output by writing the result.

*Note: The equivalent decimal value of the character 'A' is 65.

```
char str[] = { 'A', 'B', 'C', 'D', 'E', 'F' };
char *p = &str;
while (*p < str[0]+5)
{
    printf("%d ", *p++);
}
```

Question 3 (20 marks)

Predict the output of the source code (write the answer) and illustrate (analyse) the codes for any given input. What is the operation of the program?

```
double Question(int arg,...)
{
    double num=0;
    int a=0;
    va_list ap;
    va_start(ap, arg);
    for (int i =1; i <arg; i++)
        if ((a = va_arg(ap, int)) > 0)
            num += !(a%2) ? (double)pow(a,2) : (double)sqrt(a);
    va_end(ap);
    return num;
}
```

```c
int main()
{
    printf("The result is: %.2f", Question(6,1,2,3,4,5,6));
    return 0;
}
```

Question 4 (10 marks)

Predict the output of the source code (write the answer) and illustrate (analyse) the codes for any given input. What is the operation of the program?

```c
int fun(int num[],int n)
{
  int x;
  if(n == 1)
    return num[0];
  else
    x = fun(num, n-1);
  if(x > num[n-1])
    return x;
  else
    return num[n-1];
}


int main()
{

  int myArray[] = {10, 20, 30, 40, 50};
  printf(" %d ", fun(myArray, sizeof(myArray)/sizeof(myArray[0])));

  return 0;
}
```

Section B (50 marks)

Write Code

Question 1 (10 marks)

Write a function to calculate the factorial of a number using a pointer variable that is passed to the function (pass by reference). You are required to print the same pointer variable that refer to the factorial number.

Question 2 (10 marks)

Write a C program to calculate the value of variable x from the infinite series below. Please input a number from keyboard.

$$x = 1 + \frac{2}{3} + \frac{4}{5} + \frac{6}{7} + \cdots$$

*Note that the taken number from input should be an odd number.

Examples of result are shown in Figure 1, 2 and 3. Please solve the problem with a returning function using pass-by-value, and follow the input and output format of the results as shown below.

```
Enter a number: 3
The result is: 1.666667
```

Figure 1

```
Enter a number: 5
The result is: 2.466667
```

Figure 2

```
Enter a number: 7
The result is: 3.323810
```

Figure 3

Question 3 (10 marks)

Write a program to have the following function: Input 10 capital letters from keyboard, and show how many times of letter "A" you have keyed in on your monitor. Please solve the problem with function using pass-by-value. Please follow the input and output format of the result as shown below.

```
Please key in 10  Capital letters:
ABCDIHJGAE
sum=2
```

Question 4 (20 marks)

Suppose you own a car shop and repair four (4) cars every day. Write a program to input the car's information such as name of the car, weight, full fuel tank capacity, total mileage. Calculate the average of weight, full fuel tank capacity, mileage for all the cars in a day. Make sure to employ Array of struct in your program. An example of inputs and output is shown as follows:

```
Enter name: BMW
Enter weigth: 1500
Enter mileage: 400
Enter Fuel Capacity: 60

Enter Number2 car's information:
Enter name: Benz
Enter weigth: 1800
Enter mileage: 350
Enter Fuel Capacity: 60

Enter Number3 car's information:
Enter name: Toyota
Enter weigth: 1100
Enter mileage: 450
Enter Fuel Capacity: 35

Enter Number4 car's information:
Enter name: Nissan
Enter weigth: 1200
Enter mileage: 500
Enter Fuel Capacity: 35

The average of 3 features for all cars are:
The averages of mileage, weigth and FuelCapacity are respectively: 425.00 1400.00 47.50
```

- END OF PAPER -